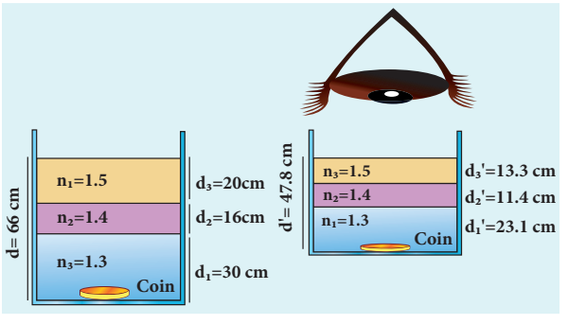
EXAMPLE 6.8

A coin is at the bottom of a trough containing three immiscible liquids of refractive indices 1.3, 1.4 and 1.5 poured one above the other of heights 30 cm, 16 cm, and 20 cm respectively. What is the apparent depth at which the coin appears to be when seen from air medium outside? In which medium the coin will appear?

Solution

When seen from (air medium) on top, the coin will still appear to be at the bottom with each medium appearing to have shrunk with respect to the air medium outside. This situation is illustrated below.



The equations for apparent depth for each medium is,

$$d_1' = \frac{d_1}{n_1}; \quad d_2' = \frac{d_2}{n_2}; \quad d_3' = \frac{d_3}{n_3}$$

$$d' = d_1' + d_2' + d_3' = \frac{d_1}{n_1} + \frac{d_2}{n_2} + \frac{d_3}{n_3}$$

$$d' = \frac{30}{1.3} + \frac{16}{1.4} + \frac{20}{1.5} = 23.1 + 11.4 + 13.3$$

$$d' = 47.8 \text{ cm}$$

6.4.6 Critical angle and total internal reflection

When a ray passes from an optically denser medium to an optically rarer medium, it bends away from the normal. Because of this, the angle of refraction r in the rarer medium is greater than the corresponding angle of incidence i in the denser medium. As angle of incidence i is gradually increased, r rapidly increases and at a certain stage r becomes 90° and the refracted ray will be gracing the boundary. **The angle of incidence in the denser medium for which the angle of refraction is 90° (or) the refracted ray graces the boundary between the two media is called *critical angle* i_c.**

If the angle of incidence in the denser medium is increased beyond the critical angle, there is no refraction possible in to the rarer medium. **For any angle of incidence greater than the critical angle, the entire light is reflected back into the denser medium itself. This phenomenon is called *total internal reflection*.** These are shown in Figure 6.21.

The two conditions for total internal reflection to take place are,

(i) light must travel from denser to rarer medium,

(ii) angle of incidence in the denser medium must be greater than critical angle ($i > i_c$).

For critical angle of incidence, the Snell's law in the product form, equation (6.19) becomes,

$$n_1 \sin i_c = n_2 \sin 90^\circ \qquad (6.29)$$

$$n_1 \sin i_c = n_2 \qquad \because \sin 90^\circ = 1$$

$$\sin i_c = \frac{n_2}{n_1} \qquad (6.30)$$

Here, $n_1 > n_2$

If the rarer medium is air, then its refractive index n_2 is 1, (n_2=1) and the refractive index of the denser medium n_1 is taken as n itself, (n_1=n) then,

$$\sin i_c = \frac{1}{n} \text{ (or) } i_c = \sin^{-1}\left(\frac{1}{n}\right) \qquad (6.31)$$

The critical angle i_c depends on the refractive index n of the medium. Table 6.3 shows the refractive index and the critical angle for different materials.

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